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# FRAUD RISK INVESTIGATION & TRANSACTION ANALYSIS

## Excel Portfolio Case Study | Synthetic Data

Tanya McGrury | Fraud and Risk Analysis Portfolio

[VIEW EXCEL WORKBOOK](#)

### PROJECT OVERVIEW

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This project began with an anomaly in a customer's transaction activity over a three-day period. Transaction frequency, dollar amounts, and geographic activity differed significantly from the customer's historical behavior, indicating that the activity warranted additional review.

I analyzed the transaction data against the customer's historical baseline, identified multiple risk indicators, and followed emerging patterns across transactions, merchants, payment terminals, and related activity. The investigation ultimately uncovered additional connections that supported escalating the case for further investigation, while distinguishing suspicious indicators from findings that could actually be supported by the available evidence.

### PROJECT AT A GLANCE

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|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| <b>TOOL</b><br>Microsoft Excel                                                                                     | <b>ANALYSIS</b><br>Transaction Analysis<br>Behavioral Baselines<br>Anomaly Detection<br>Risk Scoring   |
| <b>INVESTIGATIVE WORK</b><br>Pattern Analysis<br>Evidence Correlation<br>Case Escalation<br>Recommended Next Steps | <b>DELIVERABLES</b><br>Fraud-Risk Screening Model<br>Investigation Dashboard<br>Case Escalation Report |

#### SYNTHETIC DATA DISCLOSURE

All customer, employee, merchant, and transaction information used in this project is synthetic and was created solely for educational and portfolio demonstration purposes. No real customer or confidential business information is included.

# INVESTIGATION DASHBOARD

Initial Fraud Risk Review | Customer C104 | September 1–3, 2026

| SUSPICIOUS TRANSACTION INVESTIGATION                                                                                                                                                                                                                                             |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|---------------------|--------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|----------------------|-----------|--|--|
| Initial Fraud Risk Review   Customer C104   September 1-3, 2026                                                                                                                                                                                                                  |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
| Transactions Under Review                                                                                                                                                                                                                                                        |  |  |                     | Total Value Under Review |  |                                                                                                                                                                                                                                                                                                                             |  | Unique States        |           |  |  |
| 7                                                                                                                                                                                                                                                                                |  |  |                     | \$10,060.49              |  |                                                                                                                                                                                                                                                                                                                             |  | 3                    |           |  |  |
| Unique Devices                                                                                                                                                                                                                                                                   |  |  |                     | Review Period            |  |                                                                                                                                                                                                                                                                                                                             |  | Risk Score Breakdown |           |  |  |
| 3                                                                                                                                                                                                                                                                                |  |  |                     | 9/1/2026 - 9/3/2026      |  |                                                                                                                                                                                                                                                                                                                             |  | 5 at 4/4   2 at 3/4  |           |  |  |
| BEHAVIORAL COMPARISON   HISTORICAL VS. REVIEW PERIOD                                                                                                                                                                                                                             |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
| Metric                                                                                                                                                                                                                                                                           |  |  | Historical Activity |                          |  | Review Period                                                                                                                                                                                                                                                                                                               |  |                      | Change    |  |  |
| Average Transaction Amount                                                                                                                                                                                                                                                       |  |  | \$115.15            |                          |  | \$1,437.21                                                                                                                                                                                                                                                                                                                  |  |                      | +1148.18% |  |  |
| Unique States                                                                                                                                                                                                                                                                    |  |  | 1                   |                          |  | 3                                                                                                                                                                                                                                                                                                                           |  |                      | 1 → 3     |  |  |
| Unique Devices                                                                                                                                                                                                                                                                   |  |  | 1                   |                          |  | 3                                                                                                                                                                                                                                                                                                                           |  |                      | 1 → 3     |  |  |
| Transactions Per Day                                                                                                                                                                                                                                                             |  |  | 0.36                |                          |  | 2.33                                                                                                                                                                                                                                                                                                                        |  |                      | +553.33%  |  |  |
| KEY INVESTIGATIVE FINDINGS                                                                                                                                                                                                                                                       |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
| <b>TRANSACTION VALUE</b>   Customer C104 had a significant change in transaction behavior over a three-day period, going from a historical average of \$115.15 per transaction to \$1,437.21 per transaction, resulting in a 1,148.18% increase in average transaction amount.   |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
| <b>TRANSACTION FREQUENCY</b>   Transaction frequency also increased from 0.36 transactions per day to 2.33 transactions per day during the three-day review period, representing a 553.33% increase over the historical transaction rate.                                        |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
| <b>DEVICE &amp; GEOGRAPHY</b>   Customer C104's transactions historically originated from one device ID in one state. During the three-day review period, transactions originated from three separate device IDs across three unique states.                                     |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
| <b>ASSESSMENT</b>   Based on the significant increase in transaction value, geographic variation, transaction frequency, and use of multiple devices over the three-day review period, the activity warrants additional investigation to determine whether the transactions were |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
| INVESTIGATIVE CONCLUSION & RECOMMENDED NEXT STEPS                                                                                                                                                                                                                                |  |  |                     |                          |  |                                                                                                                                                                                                                                                                                                                             |  |                      |           |  |  |
| <b>CUSTOMER VERIFICATION</b>   Contact Customer C104 to verify whether the transactions were authorized and whether the customer remains in possession of the payment card.                                                                                                      |  |  |                     |                          |  | <b>INTERNAL REVIEW</b>   Independently review the activity associated with the additional device IDs and geographic locations, along with available transaction history, to identify any further patterns or connections. Use the combined findings to determine whether the activity is legitimate or requires escalation. |  |                      |           |  |  |

**Figure 1.** Excel investigation dashboard summarizing transaction activity, behavioral deviations, risk indicators, and recommended investigative actions for Customer C104.

Synthetic portfolio data | No real customer or confidential business information

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# CASE ESCALATION

## Evidence Review | Investigative Assessment | Recommended Next Steps

### TERMINAL ANOMALY

Merchant B terminal D301 showed a significant increase in unauthorized transaction activity, with a substantially higher dispute rate than the merchant's other terminals during the review period.

### EMPLOYEE ASSOCIATION

E17 was logged into D301 during 15 of 18 transactions later reported as unauthorized and was visually confirmed operating the terminal for 13. Four disputed transactions were manually entered while E17 operated D301 with no customer visible.

### COMMON COMPROMISE POINT

Sixteen of 17 identified customers who swiped their cards at Merchant C terminal D775 subsequently reported unauthorized activity. An unauthorized component capable of capturing magnetic-stripe data was later discovered inside the terminal.

### DOCUMENTED CONNECTION

E43 transmitted payment-card information to E17. Several entries corresponded to customer accounts previously used at D775 and subsequently used in unauthorized transactions at D301.

## INVESTIGATIVE ASSESSMENT

Evidence supports a potential connection between compromised payment-card information at Merchant C terminal D775 and subsequent unauthorized transactions processed at Merchant B terminal D301. Documented communications also show E43 transmitting payment-card information to E17, with several accounts corresponding to customers whose cards were previously used at D775 and later used in unauthorized transactions at D301.

**The available evidence did not yet establish who installed the unauthorized component, exactly how E43 obtained the payment information, or the full extent of either employee's involvement. Further investigation was warranted.**

### RECOMMENDED NEXT STEPS

- 1. Terminal Compromise Review**  
Determine when and how the unauthorized component was installed in D775.
- 2. Payment-Data Transfer Review**  
Determine how E43 obtained the payment-card information transmitted to E17.
- 3. Employee Connection Review**  
Further evaluate the documented relationship and relevant interactions between E17 and E43.
- 4. Evidence Correlation & Timeline**  
Correlate terminal activity, employee activity, communications, video, and unauthorized transactions to identify evidence that supports **or contradicts** the suspected connection.

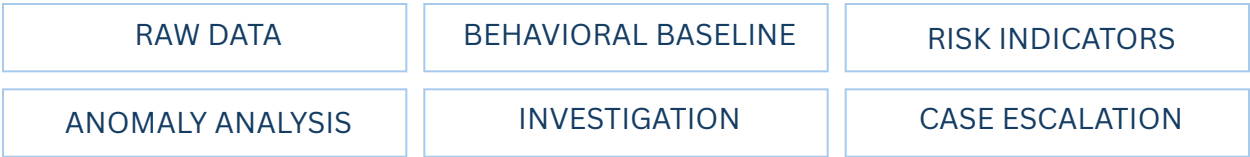
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# METHODOLOGY & SKILLS DEMONSTRATED

From Raw Transaction Data to Evidence-Based Escalation

## ANALYTICAL WORKFLOW



Analyzed synthetic transaction data to establish normal customer behavior, identify deviations across transaction value, frequency, geography, and terminal activity, and investigate emerging patterns. Findings were tested against additional evidence before determining whether escalation was warranted.

## EXCEL & DATA ANALYSIS

### EXCEL TECHNIQUES

- AVERAGEIFS and COUNTIFS
- FILTER and UNIQUE
- Dynamic arrays and spill ranges
- Cross-sheet calculations
- Date-based criteria
- Percentage-change calculations
- Excel tables and structured data
- Charts and data visualization

### ANALYTICAL TECHNIQUES

- Historical behavioral baselining
- Transaction-frequency analysis
- Multi-factor risk indicators
- Anomaly detection
- Comparative analysis
- Pattern identification
- Evidence correlation
- Risk scoring

## INVESTIGATIVE JUDGMENT

### QUESTIONED ASSUMPTIONS

Validated what “Device ID” represented rather than continuing analysis based on an incorrect interpretation of the field.

### TESTED ALTERNATIVE EXPLANATIONS

Used comparison groups, customer verification, historical activity, and additional evidence to distinguish suspicious patterns from potentially legitimate behavior.

### KEPT CONCLUSIONS EVIDENCE-BASED

Separated indicators, associations, and corroborating evidence from facts that had not yet been established.

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## PROJECT OUTPUT

**DELIVERABLES:** Excel fraud-risk screening model • Investigation dashboard • Behavioral comparison analysis • Case escalation report • Recommended investigative actions

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